

Mukul Pareek, marketing program manager, Keysight Technologies says, "Although all other interfaces have been there since quite some time, LXI Ethernet and USB interface are the latest and are widely-acceptable with the new type of PPS."

## Different output formats

After interfaces, other features associated with a PPS are output formats.

**AC + DC.** Do you only want AC output or are you also looking forward to other output forms like DC and AC + DC with an offset? If so, there are different types of PPSs today that can allow you to get different outputs based on test requirements.

Even though wave generation is the task of a wave generator, some PPSs have added functionalities that allow you to generate advanced waveforms.

**Arbitrary waveform.** These are custom-defined waveforms with either repetitive or single-wave shots. Some of these PPSs also have the ability to generate waveforms based on 1024 data points.

**Harmonic waveform.** A harmonic waveform is one whose frequency is a multiple of fundamental frequency. Some PPSs can generate custom-defined waveforms that are based on harmonic content.

**Transient generation.** Another option that some PPSs have is the ability to generate custom-defined frequency fluctuations (like sweep, swag, surge or drop-out) and voltage.

All PPSs have specific measurement options like voltage, current or power, but a few of them also have a harmonic analyser built-in to measure the Fund-50th harmonic and percentage of total harmonic distortion (THD).

## Technologies and features to look for

If you want to categorise a PPS, you have plenty of options to do so. But the most important thing to identify is the format in which it is to be categorised.



NI PXI-4110- programmable DC power supply



Rigol DP800 series programmable power supply

## Compliance test options

There are a number of compliance test options with which a PPS associates itself.

1. MIL Std. 704
2. RTCA DO 160
3. Boeing B787
4. Airbus AMD
5. Airbus ABD
6. IEC61000-4-13
7. IEC61000-3-2
8. IEC61000-3-3
9. IEC61000-3-11
10. IEC61000-3-12

A PPS can play two different roles at two different times. It can act as a source at one instance and as a sink at the other. This mode of operation is automatic, depending upon the type of application it is being used with. To explain this better, Chinmay Anand Mishra, technical marketing engineer, National Instruments says, "A PPS can be categorised as per the quadrant it is working in, that is, if it

is working as a current source or current sink."

Now, since this type of categorisation is dependent on the type of application, I will restrict myself to new technologies and features associated to categorise PPSs.

## Flexibility in voltage and current selection.

Although a PPS can be used in many applications like semiconductor fabrication, X-ray generators and crystal-growth monitoring, its main application is in automated equipment testing. While

testing, engineers need wide ranges of voltage and high currents. PPSs available till now had the maximum voltage and current settings that were not required together. Bench-top power supplies have a fixed current that results in reduction of power supplied with output voltage. New power supplies with wider voltage and current ranges are an added advantage.

## Built-in power-factor correction.

Power saving is an important parameter to be considered, especially in testing and fabrication requirements where energy losses are usual, and a device with high power factor (PF) is accountable for less energy loss. A PPS with a built-in PF corrector or controller (PFC) makes sure that its PF is maintained between 0.95 and 0.99. All power supplies sent to Europe have a mandatory built-in PFC.

**Multiple outputs.** Older PPSs were able to provide you one output at a time that could be changed with the type of input. However, new PPSs that have come up either have two completely-independent outputs, sometimes three or even four different outputs, of equal power. PPSs with these many output options have independent on/off switches to control the operations. Such outputs operate at constant current and constant voltage mode, with mode indication and automatic crossover.